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Tsi12C

Tenkasi District

Common Quarterly Examination - September 2025

25-09-25.

Standard 12

CHEMISTRY

Time Allowed: 3.00 Hours Maximum Marks: 70

PART - A

I Choose the best answer: **15×1=15**

1) Considering Ellingham diagram, which of the following metals can be used to reduce alumina?
a) Fe b) Cu c) Mg d) Zn

2) Semi conductor materials are refine by
a) Mond process b) zone refining c) Distillation d) Liquation

3) The actinoid elements which show the highest oxidation state of +7 are
a) Es, No, Lr b) U, Th, Md c) U, Fm, Th d) Np, Pu, Am

4) Among the following the correct order of acidity is
a) $HClO < HClO_2 < HClO_3 < HClO_4$ b) $HClO_2 < HClO < HClO_3 < HClO_4$
c) $HClO_4 < HClO_2 < HClO < HClO_3$ d) $HClO_3 < HClO_4 < HClO_2 < HClO$

5) The geometry at which carbon atom in diamond are bonded to each other is
a) hexagonal b) octahedral c) tetrahedral d) none of these

6) The total number of tetrahedral voids in the face centred unit cell is
a) 6 b) 8 c) 10 d) 12

7) If 'a' stands for the edge length of the cubic system, sc, bcc and fcc. Then the ratio of radii of spheres in these will be respectively
a) $\left(\frac{1}{2}a : \sqrt{3}a : \frac{1}{\sqrt{2}}a\right)$ b) $\left(\frac{1}{2}a : \frac{\sqrt{3}}{4}a : \frac{1}{2\sqrt{2}}a\right)$
c) $\left(\sqrt{1}a : \sqrt{3}a : \sqrt{2}a\right)$ d) $\left(\frac{1}{2}a : \frac{\sqrt{3}}{2}a : \frac{\sqrt{2}}{2}a\right)$

8) **Match items in Column - I with the items of Column-II and assign the correct code:**

Column - I	Column - II
A) Borazole	1. $B(OH)_3$
B) Boric acid	2. $B_3N_3H_6$
C) Quartz	3. $Na_2[B_4O_5(OH)_4].8H_2O$
D) Borax	4. SiO_2

	A	B	C	D		A	B	C	D
a)	2	1	4	3	b)	1	2	4	3
c)	1	2	4	3	d)	None of these			

9) What is the activation energy for a reaction if its rate doubles when the temperature is raised from 200k to 400k? ($R = 8.314\ JK^{-1}\ mol^{-1}$)
a) 234.65 KJ mol^{-1} b) 434.65 KJ mol^{-1} c) 2.305 KJ mol^{-1} d) 334.65 J mol^{-1}

10) $H_2PO_4^-$ the conjugate base of
a) HPO_4^{2-} b) H_3PO_4 c) P_2O_5 d) PO_4^{3-}

11) **Assertion :** Phenol is more reactive than benzene towards electrophilic substitution reaction.
Reason : In the case of phenol, the intermediate arenium ion is more stabilized by resonance.
a) Both assertion and reason are false.
b) Assertion is true but reason is false.
c) If both assertion and reason are true but reason is not the correct explanation of assertion.
d) If both assertion and reason are true and reason is the correct explanation of assertion.

12) The major product obbtained when phenol reacts with con.H_2SO_4 at 280K is
a) Salicylic acid b) Picric acid
c) o-phenol sulphonic acid d) p - phenol sulphonic acid

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13) $CH_3Br \xrightarrow{KCN} (A) \xrightarrow{H_2O} (B) \xrightarrow{PCl_5} (C)$ product "C" is
a) Chloro acetic acid b) Acetyl chloride
c) α-chloro cyano ethanoic acid d) none of these

14) The IUPAC name of $CH_2 = CH - CHO$ is
a) ethanal b) but - 2 - enal c) prop - 2 - enal d) but - 1 - enal

15) Which of these is not likely to act as Lewis base?
a) F^- b) CO c) PF_3 d) BF_3

PART - B

II. Answer any SIX questions: (Q.No. 24 is compulsory) 6×2=12

16) Which type of ores can be concentrated by froth floatation method? Give two examples for such ores.
17) Give the uses of silicones.
18) How is bleaching powder prepared?
19) What are interstitial compounds?
20) Calculate the number of atoms in an FCC unit cell.
21) Give any two difference between order and molecularity of a reaction.
22) Define solubility product.
23) How is phenolphthalein prepared?
24) What happens when ammonia is treated with benzaldehyde?

PART - C

III. Answer any SIX questions. Question No. 33 is compulsory: 6×3=18

25) Describe the role of the following in the process mentioned.
i) Silica in the extraction of copper.
ii) Cryolite in the extraction of aluminium.
26) Write a note on Fishe tropsch synthesis.
27) Give the uses of helium.
28) Describe the variable oxidation state of 3d series elements.
29) Aluminium crystallises in a cubic close packed structure. Its metallic radius is 125 pm. Calculate the edge length of unit cell.
30) Derive integrated rate law for a zero order reaction A → Product.
31) Write Schotten - Baumann reaction.
32) Explain transesterification reaction.
33) Derive the relation between pH and pOH.

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PART - D

IV. Answer all the questions: 5×5=25

34) a) i) Describe a method for refining nickel. **(3m)**
ii) Give the limitations of Ellingham diagram. **(2m)**

(OR)

b) i) How will you identify borate radical? **(2m)**
ii) Explain the preparation of Potash alum. **(3m)**

35) a) i) Give a reason to support that sulphuric acid is a dehydrating agent. **(3m)**
ii) Write the molecular formula and draw the structure of nitric acid and phosphoric acid. **(2m)**

(OR)

b) Write the differences between lanthanoids and actinoids. **(5m)**

36) a) Explain Schottky and Frenkel defects. **(5m)**

(OR)

b) Explain briefly the collision theory of bio molecular reactions. **(5m)**

37) a) Derive Henderson - Hasselbalch equation. **(5m)**

(OR)

b) i) Explain Lucas Test of differentiating three types of alcohols. **(3m)**
ii) Write note on Williamson ether synthesis. **(2m)**

38) a) Write a note on (i) Riemer - Tiemann reaction (ii) Perkin's reaction (iii) Clemmensen reduction. **(5m)**

(OR)

b) Explain the mechanism of Cannizzaro reaction.

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